

Progress Test 1.C — Elec-S1

1.C · Circuits and two-terminal components

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the power $P = 2.5 \times 10^{-2}$ kW into milliwatts. Give the exact result, without rounding.

$P = 2.5 \times 10^{-2} \times 10^3 \text{ W} = 25 \text{ W} = 25 \times 10^3 \text{ mW}$	$P = 2.5 \times 10^4 \text{ mW}$
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Marking

①	
0	1

Marking scheme

① 1 pt $2.5 \times 10^4 \text{ mW}$ (or 25 000 mW), unit written correctly.

Exercise 2. Electrical power — definition

(5 marks)

A two-terminal component, studied in the receiver convention, has a voltage $u(t)$ across it and carries a current $i(t)$. Draw the receiver convention and give the power $p(t)$ received by the component, with its SI unit expressed in terms of volts and amperes. Then give the energy it receives between t_1 and t_2 , and its expression in the DC regime ($u = u_0$ and $i = i_0$ constant during Δt).

	Receiver convention: the arrows of u and i point in opposite directions. Power received: $p(t) = u(t) i(t)$ ($p > 0$ for a receiver) Unit: the watt, $1 \text{ W} = 1 \text{ V} \times 1 \text{ A}$. Energy received: $E = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} u(t) i(t) dt$ (in joules) DC regime: $E = u_0 i_0 \Delta t$
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Marking

①		
0	1	2

②		
0	1	2

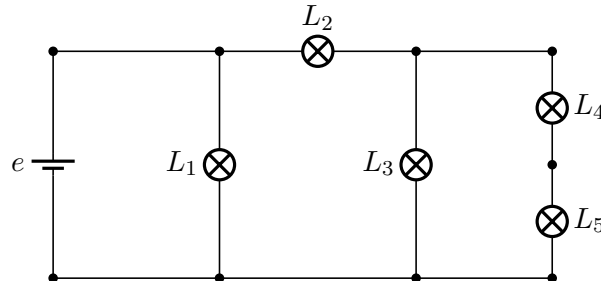
③	
0	1

Marking scheme

- ① **2 pts** Receiver convention drawn, arrows opposite (1 pt); $p = u i$ (1 pt).
- ② **2 pts** $E = \int p dt$ (1 pt); $E = u_0 i_0 \Delta t$ in the DC regime (1 pt).
- ③ **1 pt** Watt, $W = V A$.

Exercise 3. Reading a circuit diagram

(4 marks)



Base your answers on the diagram exactly as drawn, even though the same circuit could be drawn with fewer wires.

How many nodes?

☐ 2

☒ 3

☐ 4

☐ 9

Why: a node is a junction of at least three conductors, and all the points joined by wires alone are one and the same node (same potential). Distinct nodes: the top-left wire (e , L_1 , L_2), the top-right wire (L_2 , L_3 , L_4) and the whole bottom wire (e , L_1 , L_3 , L_5). The dot between L_4 and L_5 joins only two conductors, so it is not a node. “4” counts the drawn junction dots: that was the trap in Test 1 (expected answer 2, not 4).

How many branches?

☐ 4

☒ 5

☐ 6

☐ 11

Why: a branch joins two nodes: e , L_1 , L_2 , L_3 , and L_4 with L_5 in series (one single branch, since their common point is not a node).

How many two-terminal components, connecting wires included?

☐ 6

☐ 9

☒ 11

☐ 14

Why: 6 components (e , $L_1, \dots, L_5$) + 5 connecting wires (each plain segment between two dots: 1 top-left, 1 top-right, 3 at the bottom) = 11. In Test 1 the answer was $4 + 6 = 10$ on the same principle.

How many independent meshes (independent mesh equations)?

☐ 2

☒ 3

☐ 4

☐ 5

Why: $m = B - N + 1 = 5 - 3 + 1 = 3$: the three “windows” of the diagram.

Exercise 4. Energy received by a phone

(4 marks)

A phone is charged in the DC regime: the charger applies $u = 5.0 \text{ V}$ and delivers $i = 2.0 \text{ A}$ during $\Delta t = 1 \text{ h } 30 \text{ min}$. Compute the energy received by the phone, in joules, then in watt-hours.

$u = 5.0 \text{ V}$	$i = 2.0 \text{ A}$	$\Delta t = 1 \text{ h } 30 \text{ min}$
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$E = u i \Delta t$ with $\Delta t = 1.5 \text{ h} = 1.5 \times 3600 \text{ s} = 5400 \text{ s}$
$E = 5.0 \text{ V} \times 2.0 \text{ A} \times 5400 \text{ s} = 10 \text{ W} \times 5400 \text{ s}$ $E = 5.4 \times 10^4 \text{ J}$
In watt-hours: $E = 10 \text{ W} \times 1.5 \text{ h}$ $E = 15 \text{ W h}$ (check: $15 \times 3600 = 54000$)

Marking

①
0 1
□ □
②
0 1
□ □
③
0 1
□ □
④
0 1
□ □

Marking scheme

- ① 1 pt Literal expression $E = u i \Delta t$.
- ② 1 pt $\Delta t = 5400 \text{ s}$.
- ③ 1 pt $E = 5.4 \times 10^4 \text{ J}$.
- ④ 1 pt $E = 15 \text{ W h}$.

Examiner's tip

"1 h 30 min" is 1.5 h, not 1.3 h. Convert times into seconds before using SI formulas.

Your score: / 14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = / 20